



Lemon Fresh Dishwashing Liquid — Batch Sheet

Density 1.03 kg/L

Rev. Sept 2026

Sofex Soap & Chemical Industries · weigh everything, never measure by volume

STANDARD GRADE · 15% ANIONIC ACTIVE

CHARGE QUANTITIES

#	INGREDIENT	%	10 L	20 L	50 L	100 L	✓
1	Water (soft/DM) — first charge	45.0	4.64 kg	9.27 kg	23.2 kg	46.4 kg	
2	LABSA 96%	9.00	927 g	1.85 kg	4.63 kg	9.27 kg	
3	Caustic soda flakes 98% (as 30% soln)	1.15	118 g	237 g	592 g	1.18 kg	
4	SLES 70%	8.00	824 g	1.65 kg	4.12 kg	8.24 kg	
5	CAPB 30%	6.00	618 g	1.24 kg	3.09 kg	6.18 kg	
6	Cocamide MEA	1.50	155 g	309 g	773 g	1.55 kg	
7	Glycerine	1.00	103 g	206 g	515 g	1.03 kg	
8	Tetrasodium EDTA 39%	0.30	31 g	62 g	155 g	309 g	
9	Preservative	0.08	8 g	16 g	41 g	82 g	
10	Colour, 1% solution	0.05	5 g	10 g	26 g	52 g	
11	Fragrance (lemon)	0.30	31 g	62 g	155 g	309 g	
12	Citric acid (pH trim, as 20% soln)	0.10	10 g	21 g	52 g	103 g	
13	Sodium chloride (as 25% brine, to viscosity)	~2.00	~206 g	~412 g	~1.03 kg	~2.06 kg	
14	Water — final make-up to weight	bal.	2.62 kg	5.23 kg	13.1 kg	26.2 kg	
	Batch weight	100.00	10.3 kg	20.6 kg	51.5 kg	103.0 kg	
	450ml bottles produced	—	22	44	111	222	

ORDER OF WORK

- Caustic solution first.** Dissolve the flakes in their share of water (30% strength) — **flakes into water, never the reverse**. Let it cool below 35 °C.
- Charge the first water into the vessel, start the stirrer, add LABSA slowly over 10–15 min.
- Add the cooled caustic slowly over 20–30 min. **Stop at pH 7.0–7.5**. Keep below 45 °C. Stir 15 min.
- Cool below 35 °C before anything else goes in.
- SLES in portions until clear (20–30 min), then CAPB (10 min), then cocamide MEA until no specks remain.
- Glycerine, EDTA, preservative — each fully in before the next.
- Below 32 °C: colour solution, then fragrance into the vortex. Stir 15 min.
- Re-check pH; trim with citric acid solution if above 7.5.
- Brine in 0.2% steps.** Stir 10 min, cool a sample to 25 °C, read viscosity. Stop on the rising side at 1200–1800 cps.
- Final water to exact batch weight. Stir 20 min.
- Rest 8–12 hours. Run QC. Filter 100 mesh. Fill and batch-code.

PH AT 25 °C	VISCOSITY	ACTIVE MATTER	MAX TEMP	SP. GRAVITY
6.5 — 8.0	1200 — 1800 cps	≥ 15.0%	45 °C	1.020 — 1.050

Safety — every batch, no exceptions. Caustic soda and LABSA both cause serious chemical burns. Face shield over goggles, elbow-length nitrile gloves, rubber apron, closed shoes. Caustic **always into water**. Neutralisation gives off real heat — add slowly and watch the thermometer. Alkali burns feel mild at first and keep damaging tissue: flush with running water for 20 minutes and go to hospital. Never store or mix any of this near bleach.

Batch record

BATCH NO.	DATE	SIZE	OPERATOR	PH	VISCOSITY	ACTIVE %	QC SIGNATURE

Sofex Soap & Chemical Industries — internal batch sheet. Starting formulation: confirm on a 10-litre trial batch with your own raw-material lots, and run a full stability study, before scaling to production. Salt quantity is indicative — always titrate to viscosity.